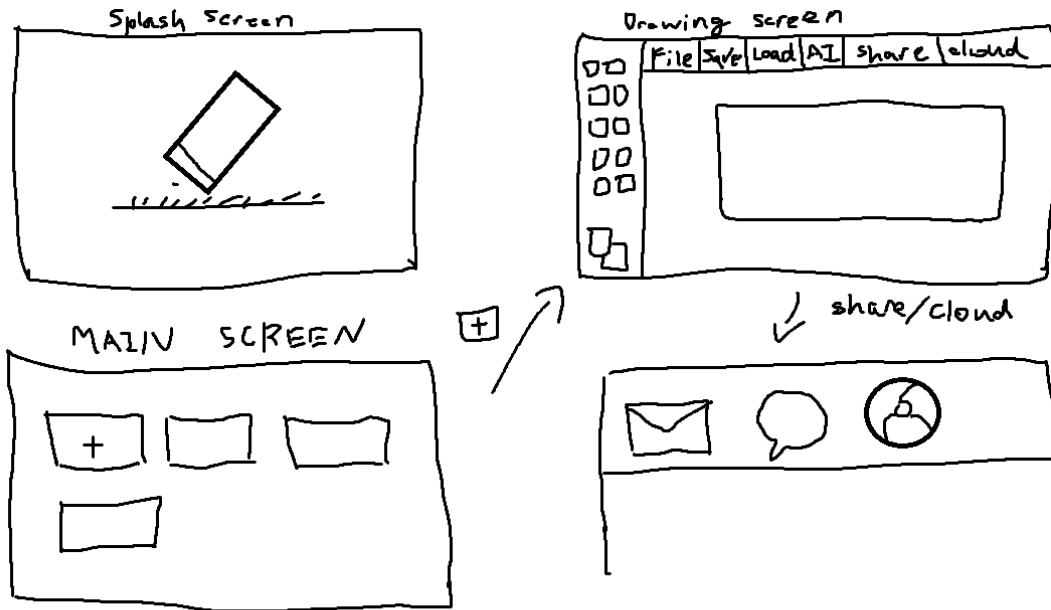


Phase 1 Project Plan

Sketches/Wireframes



To-Do List

- Unit Tests
- Splash Screen
- Main Screen/Dashboard + Open/Edit Project screen (combined?)
- Drawing Screen
- A screen to manage cloud backups or sharing functionality. (View only)
- AI-enabled image analysis
 - Pick some image recognition API like <https://cloud.google.com/vision>
 - Give the API relevant data (onClick of AI button) and display the results

(All With Implied Unit Tests)	Date/Precedence	Assigned To:	Status
Splash Screen	First/Early ▾	Chester ▾	Not Started ▾
Dashboard	First/Early ▾	Chester ▾	Not Started ▾
Drawing Screen	First/Early ▾	Zack ▾	Not Started ▾
Sharing Screen or Cloud Storage (Screen)	First/Early ▾	Roscoe ▾	Not Started ▾
AI Recognition Screen	Last/Low-Pri... ▾	Chester ▾	Not Started ▾
Research AI APIs	Last/Low-Pri... ▾	Un-Assigned ▾	Not Started ▾
Research Sharing APIs	First/Early ▾	Roscoe ▾	Not Started ▾
Research cloud storage APIs	First/Early ▾	Roscoe ▾	Not Started ▾

Task Breakdown

- Layouts/Screens (View)
 - Splash Screen: No buttons or anything to interact with. Just waiting for Main Screen to load
 - Main Screen (Dashboard). Images of past drawings in a grid and a + button that allows the user to create a new drawing and go to the drawing screen.
 - Drawing Screen: Includes many buttons that bring up tools that help the user draw what they want. Might include eraser, pencil etc. Save/Load button.
 - AI Interface (possibly as a pullup screen): Button to import image, and a button that runs the AI-powered image analysis to detect objects in the image. Output box to display list of detected objects.
 - Sharing (possibly as a pullup screen): Buttons to share drawing to other apps (like OneDrive or Discord).
- Classes/Methods (Model View)
 - AddDrawing(): Called when the + button on the dashboard is clicked
 - Save(): Called when the save button is clicked on the drawing screen. Adds the drawing to the dashboard.

- Load(): Called when an image is clicked on the dashboard or when the load button is called on the drawing screen. Loads the image in the drawing screen.
- Canvas Class
- ToolSelect(Tool tool): Called when a tool is selected in the Drawing Screen. Switches to selected tool.
- Draw(): Called when the pen tool or eraser tool is selected. Draws or erases pixels at applied locations.
- AIButton(): Called when the AI button is clicked on the drawing screen. Opens the AI Interface.
- OnLoad(): Load MostRecent(localVersion, cloudVersion)
- OnShareClick(): Pulls up sharing screen
- BackUp(): Run onClose() and periodically during editing. Pushes local to remote for files with sync turned on.
- Model
 - (Pending Definition of User/Cloud requirements) Will have files stored locally that can be loaded from or saved to.
- Sharing and Cloud Backups
 - Research sharing APIs
 - <https://developer.android.com/social-and-messaging/guides/media-sharing>
 - <https://docs.discord.com/developers/reference>
 - <https://learn.microsoft.com/en-us/graph/use-the-api>
 - Integrating user login functionality with stuff like OneDrive or DropBox
- Unit Tests
 - Checks that a new Canvas is created when + is clicked
 - Checks drawings are saved internally in some form when the user makes changes to the canvas
 - Checks drawings are saved externally when the save button is clicked
 - Checks the user is directed back to the dashboard when a back arrow/button clicked
 - Checks the splash screen is displayed
 - Checks the correct drawings are loaded to the drawing screen
 - Checks correct tools are selected

Order

- Phase 1 will mostly be setting up all the different screens in the layout (view). Starting with the splash screen and main screen will allow us to test as we go. The main screen includes buttons that lead to other screens.
- Then, once the view is set up, we can set up drawing capabilities on the drawing page and code the logic (Model View) that updates the Model.
- As we code the logic, we'll also design the unit tests so we can test while programming.
- Once all the basic logic is set up, we can focus on sharing and integrating AI into the application
- Concurrent with AI integration we can also work on sharing and cloud integration.